

E-Learning Platform Analytics using MySQL

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- analysis
- hospital
- learnersdb
 - Tables
 - Views
 - Stored Procedures
 - Functions
- library_db
- sakila
- socialmedia
- socialmediadb
- sys
- world

Administration Schemas

Information

No object selected

Query 1: session 3-SQL Clauses class co. Session 4-reference

Limit to 1000 rows

```
66 inner join purchase on learners.learner_id = purchase.learner_id
67 join courses on purchase.course_id = courses.course_id;
68
69 select full_name as learner_name, course_name, category,
70 format(quantity * unit_price, 2) as Total_amount, purchase_date
71 from learners
72 inner join purchase on learners.learner_id = purchase.learner_id
73 join courses on purchase.course_id = courses.course_id order by total_amount desc;
```

Result Grid

learner_name	course_name	category	Total_amount	purchase_date
Husna	Financial accounting	Finance	90,000.00	2025-01-05
Priya	Digital marketing	Marketing	80,000.00	2025-01-04
Gopi	Python programming	Programming	50,000.00	2025-01-01
Divya	Graphic design	Design	140,000.00	2025-01-03
Divya	Graphic design	Design	140,000.00	2025-01-06
Nagesh	Data analyst	Data science	120,000.00	2025-01-02
Priya	Python programming	Programming	100,000.00	2025-01-07

Result 7 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	20:10:34	select * from courses LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
14	22:31:31	select full_name, purchase_date from learners inner join purchase on learners.learner_id = purchase.learner_id	7 row(s) returned	0.000 sec / 0.000 sec
15	22:53:34	select full_name, course_name, purchase_date from Learners inner join purchase on learners.learner_id = purch...	7 row(s) returned	0.000 sec / 0.000 sec
16	23:21:31	select full_name as learner_name, course_name, category, quantity * unit_price as Total_amount, purchase_date	7 row(s) returned	0.000 sec / 0.000 sec
17	23:32:51	Format (quantity * unit_price, 2) as total_amount from purchase	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
18	23:45:32	select full_name as learner_name, course_name, category, format (quantity * unit_price, 2) as Total_amount, purch...	7 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

24°C Mostly cloudy

Search

ENG IN

23:45 15-08-2026

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Query 1: session 3-SQL Clauses class co. Session 4-reference

Limit to 1000 rows

```
66 inner join purchase on learners.learner_id = purchase.learner_id
67 join courses on purchase.course_id = courses.course_id;
68
69 select full_name as learner_name, course_name, category, quantity,
70 format(quantity * unit_price, 2) as Total_amount, purchase_date
71 from learners
72 inner join purchase on learners.learner_id = purchase.learner_id
73 join courses on purchase.course_id = courses.course_id order by total_amount desc;
```

Result Grid

learner_name	course_name	category	quantity	Total_amount	purchase_date
Husna	Financial accounting	Finance	1	90,000.00	2025-01-05
Priya	Digital marketing	Marketing	1	80,000.00	2025-01-04
Gopi	Python programming	Programming	1	50,000.00	2025-01-01
Divya	Graphic design	Design	2	140,000.00	2025-01-03
Divya	Graphic design	Design	2	140,000.00	2025-01-06
Nagesh	Data analyst	Data science	2	120,000.00	2025-01-02
Priya	Python programming	Programming	2	100,000.00	2025-01-07

Result 8 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	22:31:31	select full_name, purchase_date from learners inner join purchase on learners.learner_id = purchase.learner_id	7 row(s) returned	0.000 sec / 0.000 sec
15	22:53:34	select full_name, course_name, purchase_date from Learners inner join purchase on learners.learner_id = purch...	7 row(s) returned	0.000 sec / 0.000 sec
16	23:21:31	select full_name as learner_name, course_name, category, quantity * unit_price as Total_amount, purchase_date	7 row(s) returned	0.000 sec / 0.000 sec
17	23:32:51	Format (quantity * unit_price, 2) as total_amount from purchase	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
18	23:45:32	select full_name as learner_name, course_name, category, format (quantity * unit_price, 2) as Total_amount, purch...	7 row(s) returned	0.000 sec / 0.000 sec
19	23:46:19	select full_name as learner_name, course_name, category, quantity, format (quantity * unit_price, 2) as Total_amo...	7 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

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Search

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23:46 15-08-2026

MySQL Workbench

Local instance MySQL80

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Navigator: Schemas

learnersdb

Tables

Views

Stored Procedures

Functions

Showing loaded schemas only

Administration Schemas

Information

Schema: learnersdb

Create database if not exists LearnersDB;

use LearnersDB;

Create table if not exists Learners (

learner_id int primary key,

Full_name varchar(100),

Country varchar(100));

Create table if not exists Courses (

Course_id int Primary key, Course_name varchar(100), Category varchar(100), Unit_price int)

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	18:12:14	Create database if not exists LearnersDB	1 row(s) affected	0.000 sec
2	18:12:36	use LearnersDB	Error Code: 1049. Unknown database 'learnersdb'	0.000 sec
3	18:12:56	use LearnersDB	0 row(s) affected	0.016 sec
4	18:20:15	Create table if not exists Learners (learner_id int primary key; Full_name varchar(100); Country varchar(100))	0 row(s) affected	0.046 sec
5	18:24:31	Create table if not exists Courses (Course_id int Primary key; Course_name varchar(100); Category varchar(100); Unit_price int)	0 row(s) affected	0.047 sec
6	18:34:22	Create table if not exists Purchase (Purchase_id int, learner_id int, Course_id int, Quantity int, Purchase_date date)	0 row(s) affected	0.015 sec

Object Info Session

Added new script editor

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MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: Schemas

learnersdb

Tables

Views

Stored Procedures

Functions

Showing loaded schemas only

Administration Schemas

Information

No object selected

session 3-SQL Clauses class co. LearnersDB Session 4-reference

format (quantity * unit_price,2) as Total_amount,purchase_date from learners

inner join purchase on learners.learner_id = purchase.learner_id

join courses on purchase.course_id = courses.course_id

order by total_amount desc;

select full_name as learner_name,Country,sum(quantity*unit_price) as total_spending

from learners join purchase on learners.learner_id = purchase.learner_id

join courses on purchase.course_id = courses.course_id

group by learner_name,country;

Result Grid

learner_name	Country	total_spending
Gopi	India	50000
Nagesh	China	120000
Divya	UAE	280000
Priya	South Korea	180000
Husna	Japan	90000

Result 1 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
16	23:21:31	select full_name as learner_name,course_name,category,quantity * unit_price as Total_amount,purchase_date...	7 row(s) returned	0.000 sec / 0.000 sec
17	23:32:51	Format (quantity*unit_price,2) as total_amount from purchase	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
18	23:45:32	select full_name as learner_name,course_name,category, format (quantity * unit_price,2) as Total_amount,purc...	7 row(s) returned	0.000 sec / 0.000 sec
19	23:46:19	select full_name as learner_name,course_name,category,quantity, format (quantity * unit_price,2) as Total_amo...	7 row(s) returned	0.000 sec / 0.000 sec
20	00:33:51	select learner_name,Country,sum(quantity*unit_price) as total_spending from learners join purchase on learners...	Error Code: 1054. Unknown column 'learner_name' in field list	0.000 sec
21	00:34:31	select full_name as learner_name,Country,sum(quantity*unit_price) as total_spending from learners join purchas...	5 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

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Administration Schemas

Information

Schema: learnersdb

session 3-SQL Clauses class co. LearnersDB Session 4-reference

Limit to 1000 rows

```
88 FROM learners, courses, purchase
89 WHERE learners.learner_id = purchase.learner_id
90 AND purchase.course_id = courses.course_id
91 GROUP BY learners.learner_id, full_name
92 HAVING COUNT(DISTINCT category) > 1;
93
94 • select purchase_id,course_name,courses.course_id,quantity from courses
95 left join purchase on purchase.course_id = courses.course_id
96 where purchase.quantity is null;
97
```

Result Grid

purchase_id	course_name	course_id	quantity
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Result 3 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	10:48:17	Select full_name as learner_name, Purchase_id,count(distinct category) as category_count from learners.cours...	Error Code: 1054. Unknown column 'learnerfull_name' in 'group statement'	0.000 sec
4	10:48:48	Select full_name as learner_name, Purchase_id,count(distinct category) as category_count from learners.cours...	Error Code: 1055. Expression #2 of SELECT list is not in GROUP BY clause and contains nonaggregated colu...	0.000 sec
5	10:49:35	SELECT full_name AS learner_name, COUNT(DISTINCT category) AS category_count FROM learners...	1 row(s) returned	0.016 sec / 0.000 sec
6	11:58:57	select purchase_id,course_name,course_id,quantity from courses left join purchase on purchase.course_id = ...	Error Code: 1052. Column 'course_id' in field list is ambiguous	0.000 sec
7	12:00:42	select purchase_id,course_name,course_id,quantity from courses left join purchase on purchase.cours...	Error Code: 1054. Unknown column 'course course_id' in field list	0.000 sec
8	12:01:15	select purchase_id,course_name,courses.course_id,quantity from courses left join purchase on purchase.cour...	0 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

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Administration Schemas

Information

No object selected

session 3-SQL Clauses class co. LearnersDB Session 4-reference

Limit to 1000 rows

```
109 JOIN courses
110 ON purchase.course_id = courses.course_id
111 GROUP BY learner_id) AS spending);
112
113 • SELECT course_name,category,unit_price
114 FROM courses
115 WHERE unit_price > ANY (
116 SELECT unit_price
117 FROM courses WHERE Category = "Beginner");
118
```

Result Grid

course_name	category	unit_price
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courses 2 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:46:38	SELECT (full_name AS learner_name,SUM(p.quantity * c.unit_price) AS total_spending FROM learners l JOIN p...	2 row(s) returned	0.000 sec / 0.000 sec
2	20:18:26	SELECT course_name,category,unit_price FROM courses WHERE unit_price > ANY (SELECT unit_price FR...	0 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

27°C Partly cloudy

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Administration Schemas

Information

No object selected

```

156 WHEN SUM(p.quantity * c.unit_price) >= 8000 THEN 'Medium Value'
157 ELSE 'Low Value'
158 END AS spending_category
159 FROM learners l
160 JOIN purchase p
161 ON l.learner_id = p.learner_id
162 JOIN courses c
163 ON p.course_id = c.course_id
164 GROUP BY l.learner_id, l.full_name;
165

```

Result Grid

learner_name	total_spending	spending_category
Gop	50000	High Value
Nagesh	120000	High Value
Divya	280000	High Value
Priya	180000	High Value
Huma	90000	High Value

Result 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:46:38	SELECT l.full_name AS learner_name, SUM(p.quantity * c.unit_price) AS total_spending FROM learners l JOIN p...	2 row(s) returned	0.000 sec / 0.000 sec
2	20:18:26	SELECT course_name, category, unit_price FROM courses WHERE unit_price > ANY (SELECT unit_price FR...	0 row(s) returned	0.000 sec / 0.000 sec
3	20:19:41	SELECT DISTINCT category FROM courses LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
4	20:33:23	SELECT l.full_name AS learner_name, l.country, SUM(p.quantity * c.unit_price) AS total_spending FROM learners ...	0 row(s) returned	0.015 sec / 0.000 sec
5	20:35:47	WITH learner_spending AS (SELECT l.full_name AS learner_name, SUM(p.quantity * c.unit_price) AS total_spen...	5 row(s) returned	0.016 sec / 0.000 sec
6	20:36:44	SELECT l.full_name AS learner_name, SUM(p.quantity * c.unit_price) AS total_spending, CASE WHEN SUM(p.q...	5 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

27°C Partly cloudy

Search

ENG IN

20:36 16-08-2026

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Administration Schemas

Information

No object selected

```

175 COUNT(p.purchase_id) AS number_of_purchases,
176 SUM(p.quantity * c.unit_price) / COUNT(p.purchase_id) AS average_revenue_per_purchase
177 FROM courses c
178 JOIN purchase p
179 ON c.course_id = p.course_id
180 GROUP BY c.category;
181
182 SELECT *
183 FROM category_performance_view;
184

```

Result Grid

category	total_revenue	number_of_purchases	average_revenue_per_purchase
Programming	150000	2	75000.0000
Data science	120000	1	120000.0000
Design	280000	2	140000.0000
Marketing	80000	1	80000.0000
Finance	90000	1	90000.0000

category_performance_view 8 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	20:33:23	SELECT l.full_name AS learner_name, l.country, SUM(p.quantity * c.unit_price) AS total_spending FROM leame...	0 row(s) returned	0.015 sec / 0.000 sec
5	20:35:47	WITH learner_spending AS (SELECT l.full_name AS learner_name, SUM(p.quantity * c.unit_price) AS total_spe...	5 row(s) returned	0.016 sec / 0.000 sec
6	20:36:44	SELECT l.full_name AS learner_name, SUM(p.quantity * c.unit_price) AS total_spending, CASE WHEN SUM(p.q...	5 row(s) returned	0.000 sec / 0.000 sec
7	20:37:38	SELECT c.course_name, IFNULL(COUNT(p.purchase_id), 0) AS purchase_count FROM courses c LEFT JOI...	0 row(s) returned	0.015 sec / 0.000 sec
8	20:38:28	CREATE VIEW category_performance_view AS SELECT c.category, SUM(p.quantity * c.unit_price) AS total_r...	0 row(s) affected	0.031 sec
9	20:38:41	SELECT * FROM category_performance_view LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

27°C Partly cloudy

Search

ENG IN

20:38 16-08-2026